

A Literature Review of Predictive Analytics for Pre-Surgical Decision
Support in Coronary Stent Placement

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ABSTRACT

This project explores how Machine learning algorithms can interpret patient vital signs and imaging data to predict complications before PCI stent placement surgery. Doctors currently rely on their personal experience to interpret clinical reports. They use predictions made by general risk scoring systems like Syntax/TIMI to make surgical decisions. Patients solely rely on doctor's judgment when deciding whether to undergo this high-risk procedure. This project explores how predictive analytics can process large amounts of clinical data and produce personalized and accurate risk predictions for stent placement outcomes. This review examines studies published in major journals such as Nature, BMC Medical Informatics and NPI Digital Medicine. Primarily, this review shows how ML algorithms such as logistic regression, random forest and XGBoost predict PCI complications like stent under expansion and no reflow. This review explains how explainable AI tools justify predictions by showing which attribute contributes most to an accurate prediction. The review also shows that combining vital signs with imaging data increases predictive accuracy. This review identifies the steps researchers take to ensure accuracy and ethical use of AI in the clinical settings. In conclusion, this literature review proves how XAI and ML algorithms can support patient-centered and data-driven surgical decisions in coronary stent placement.

Keywords: Predictive analytics, machine learning, coronary stent placement, explainable AI, data integration, decision-support systems.

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I. INTRODUCTION

Percutaneous Coronary Intervention (PCI) is a medical procedure to help restore flow in blocked arteries carrying blood to the heart. The outcome of this surgery can vary widely from patient-to-patient based upon various factors such as age, artery size, plaque buildup and other vital conditions such as diabetes, BP and kidney functioning. There are several risks involved in this surgery such as stent under expansion, artery rupture, no reflow. Sometimes the outcome can be so critical that it might even result in the death of the patient. Traditionally, cardiologists rely on their past experience to make decisions on the need for surgery. They use general risk coding systems such as SYNTAX and TMI to explore whether the surgery can be effective. These traditional models consider only limited variables and provide general estimates instead of precise patient-centric estimates [1]. If there was a way to predict effectively the outcome of a surgery prior to happening, several patients can be benefited. If they predict high risk before surgery itself, patients can avoid this surgery and choose other life-extending alternate therapy. ML can address this gap of not having patient-specific risk prediction. It can digest vast amount of clinical and imaging data and identify hidden patterns to predict the risk involved and outcome of the surgery. When paired with explainable AI, these models apart from predicting risk can also suggest which factor contributed the most to that risk prediction. This review examines recent developments in predictive analytics for medical treatment and specifically in cardiovascular line. It explores how combining clinical and imaging data can be effective, how explainable model provide clarity and how visualization can help communication between doctor and patient. Finally, the review

also shows how effort has been taken to effectively use AI in a safe and ethical manner in the medical settings.

II. MACHINE LEARNING FOR CARDIOVASCULAR TREATMENT PREDICTION

This section explores how Machine Learning supports clinical decision making by predicting surgical risk before PCI stent placement treatment. Firstly, the study conducted by Gharaibeh in Jordan University Hospital [1] provides highly relevant evidence for this review because it focuses directly on PCI patients with calcified arteries. They used OCT scan images to train ML models like support vector machine and random forest to predict stent under expansion. This is a post-surgery complication that commonly affects stent success. The researchers first used pre-surgical data and made predictions on it. They also collected post-surgery data and compared the values to the predictions made and validated it. This study confirms that ML system can provide reliable foresight before the operation even begins. However, its narrow focus on calcified arteries limits its broader applicability across diverse PCI cases.

The study by Deng and his team offers broader clinical relevance because it analyzes 2,000 PCI patients with ST-segment elevation myocardial infarction [2]. This study uses a wider range of patient-specific variables, including cholesterol levels and angiography report readings that provides blockage size. Using this extensive data, they trained ML models such as gradient boosting and random forest to predict issue of no reflow i.e. failure of blood flow and potential death after this stent placement surgery. This model achieved a very high accuracy (AUC above 0.85). They also proved that this technique

could predict better than traditional regression model that doesn't identify non-linear relationship between patient-specific factors and the surgery outcome. This wide dataset and higher accuracy make Deng's findings more generalizable and clinically useful.

Studies outside cardiovascular care also contribute important context about the broader role of ML in clinical prediction. For example, the study conducted by Rajkomar is a proof [3]. He analyzed over 200,000 anonymous electronic health records of patients across several US hospitals. He developed a deep learning system that predicted a wide range of patient outcomes right from mortality to the likelihood of readmission. Shickel's study reinforces Rajkomar's conclusions by showing that automated ML pipelines reduce human bias and improve prediction accuracy. [4] Both of these studies show wide applicability, but its general scope makes it less directly relevant to PCI-specific complications compared to Gharaibeh and Deng.

The research by Tomašev and the UK National Health Service highlights the value of real-time prediction in hospital settings [5]. They developed a deep learning model to continuously monitor kidney failure patients who were hospitalized. They predicted acute kidney condition up to 48 hours in advance. They demonstrated how early warnings can change treatment outcomes. This study strengthens the argument that ML can support continuous monitoring, but it contributes less directly to PCI risk prediction because it focuses on kidney failure rather than cardiovascular surgery.

Across all these studies, machine learning consistently improves predictive accuracy in clinical settings. However, researchers are constantly worried about the two major concerns that limits its usefulness. First, the medical datasets used for these predictions lack standardization across hospitals and devices. Second, these ML models produce

outputs that clinicians find difficult to interpret. Addressing these issues is essential to ensure that ML is effectively used in medical practice.

III. INTEGRATING CLINICAL AND IMAGING DATA

This section of the review explores how identifying data variables from imaging data like medical scans when combined with traditional patient data like vitals can help improve the accuracy of AI models.

To support this, research conducted by Lundervold from University of Bergen in Norway was studied [6]. They demonstrated that neural networks with no other medical intervention or information provided could analyze MRI scans and identify patterns that give information about how patients respond to medical treatment. This study demonstrates that imaging alone can reveal patterns that reflect how patients respond to treatment.

The study by Shen supports Lundervold's findings by using deep learning to study scans of patients having cancer [7]. They found that AI can identify details in the medical scan that even experts may overlook. Shen's study reinforces the value of imaging data, proving that similar capabilities can be used in cardiovascular prediction.

The study by Li contributes stronger relevance because it applies deep learning to cardiovascular imaging specifically [8]. They used the same technique of deep learning to locate blockages in blood vessels. Raissi-Dehkordi's research also shows how deep learning can improve cardiovascular ultrasound scans [9]. This research specifically

measures the size of the heart chamber accurately. Though some of the research covered here focus mainly on study of imaging data rather than combining it with patient vital data, it is helpful in showing how the usage AI in medical imaging is effective.

Collectively, all these studies have shown that with imaging data combined the AI models tend to capture the total complexity of the disease even more accurately and effectively. In all these research, public data set like MIMIC-IV has been used for the research. This contains de-identified patient data that is vast and hence helpful in large-scale training. These data sets also have their own disadvantages such as having different data formats and missing information. All this has to be addressed in our study to effectively use the existing data to make valuable predictions.

IV. EXPLAINABLE AI IN MEDICAL PREDICTION

This section focuses on the concept of explainable AI and its benefits in creating clinical trust in the predictive model. Having an accurate model is not just enough. The results should be easily interpretable, especially in widely skilled medical teams. XAI helps achieve this simplicity.

Explainable AI has many related frameworks. One such, the SHAP (Sharpley Additive Explanations), was introduced by Lundberg and Lee[10]. This framework shows how much each factor influences a model's overall prediction. This model shows both global explanation for the overall model and patient-level inferences. Holzinger [11] emphasized in his paper that this interpretation is most essentially needed to ensure wide acceptance of AI and also to meet the regulatory needs.

Another framework in XAI is the LIME (Local Interpretable Model Agnostic Explanation), developed by Riberio[12]. In this framework, apart from the complex model, a simplified, easy-to-understand version is developed. This version can be used to explain why a prediction was made for each single case. By combining both SHAP and LIME, doctors and patients can see which factors contribute to a risk prediction. Although these methods require extra computation and time, they remain essential for building trustworthy and clinically usable AI systems.

V. VISUALIZATION TOOLS FOR CLINICAL COMMUNICATION

This section discusses how visualization tools can help improve communication between medical staff and patients.

Interactive dashboards allow clinicians to enter patient data and instantly view the associated risk predictions. These systems often use simple color codes, with green showing low risk, yellow for moderate, and red for high. SHAP or LIME plots when integrated into these dashboards can highlight the exact features (like high cholesterol or artery diameter) that most contributed to the predicted outcome [1]. Studies prove that patients remember information better when presented visually rather than being mentioned verbally. These dashboards can also be helpful for care coordination between medical staff (cardiologists, technicians, and nurses) during treatment planning.

VI. ETHICAL AND IMPLEMENTATION CONSIDERATIONS

This section reviews the ethical and implementation challenges commonly observed while using AI in healthcare.

While AI can improve precision and personalization, it also comes with its own complex ethical concerns. Models must use de-identified data and comply with HIPAA standards to protect patient confidentiality [3].

It is a must to confirm that the model works similar for individuals across all geographies, age groups and genders. Holzinger in his paper mentions that failing to test for these factors could have serious repercussions especially in healthcare [11]. The same concern was stressed upon by Tomaev in his research [5]. He emphasized that the Model trained in one hospital should perform reliably in a different hospital as well. Patient population, equipment and medical expertise can differ across each location. He mentions that with frequent model re-checking and updates this issue must be fixed and be made available to work in a similar manner across diverse settings. Most importantly doctors and nurses should be provided proper training to interpret and use these new AI tools.

VII. CONCLUSION

This literature review proves that Machine Learning (ML) with Explainable Artificial Intelligence (XAI) can improve how doctors evaluate surgical risks. By integrating patient data and imaging information, deep learning-based AI models have high potential to generate accurate and patient centric predictions. XAI tools like SHAP and LIME can make doctors clearly see and explain to patients which characteristic most influenced a prediction. Additionally interactive dashboards enable ease of use and improves communication between the doctor and the patient. Overall, Predictive analytics help reduce personal bias and improve accuracy. When used alongside a doctor's expertise it is most promising to improve medical decision-making

However beneficial, there are certain challenges associated with the use of AI. These include varying data standards across hospitals and the need for models to work consistently across all demographics. Extensive training and testing can help overcome these drawbacks.

REFERENCES

1	Gharaibeh, A. A., Alquran, M. H., & Alqudah, A. M. (2023). <i>Prediction of stent under-expansion in calcified coronary arteries using machine learning on intravascular OCT</i> . <i>Scientific Reports</i> , 13, 18417. https://www.nature.com/articles/s41598-023-44610-9
2	Deng, D., Sun, Y., Zhou, J., et al. (2022). <i>Machine learning to predict no-reflow and in-hospital mortality in patients with ST-segment elevation myocardial infarction undergoing primary PCI</i> . <i>BMC Medical Informatics and Decision Making</i> , 22. https://link.springer.com/content/pdf/10.1186/s12911-022-01853-2
3	Rajkomar, H., Oren, E., Chen, K., et al. (2018). <i>Scalable and accurate deep learning for electronic health records</i> . <i>npj Digital Medicine</i> , 1, 18. https://www.nature.com/articles/s41746-018-0029-1
4	Shickel, J., Tighe, P., Bihorac, A., & Rashidi, P. (2017). <i>Deep EHR: A survey of recent advances in deep learning techniques for electronic health record analysis</i> . <i>arXiv preprint arXiv:1706.03446</i> . https://arxiv.org/pdf/1706.03446.pdf
5	Tomašev, A., Glorot, X., Rae, J. W., et al. (2019). <i>A clinically applicable approach to continuous prediction of future acute kidney injury</i> . <i>Nature</i> , 572, 116–119. https://doi.org/10.1038/s41586-019-1390-1
6	Lundervold, J., & Lundervold, A. (2019). <i>An overview of deep learning in medical imaging focusing on MRI</i> . <i>Zeitschrift für Medizinische Physik</i> , 29(2), 102–127. https://arxiv.org/pdf/1811.10052.pdf
7	Shen, S., Han, M., Wei, L., Guo, J., & Hao, Y. (2020). <i>Deep learning for medical image analysis in cancer diagnosis</i> . <i>Frontiers in Oncology</i> , 10, 565. https://www.frontiersin.org/articles/10.3389/fonc.2020.00565/full
8	Li, X., Liang, J., Huang, C., & He, Y. (2020). <i>Deep learning-enabled medical computer vision</i> . <i>npj Digital Medicine</i> , 3. https://www.nature.com/articles/s41746-020-00376-2
9	Raissi-Dehkordi, N., Raissi-Dehkordi, N., & Xu, B. (2025). <i>Contemporary applications of artificial intelligence and machine learning in echocardiography</i> . <i>npj Cardiovascular Health</i> . https://www.nature.com/articles/s44325-025-00064-8
10	Lundberg, S. M., & Lee, S.-I. (2017). <i>A unified approach to interpreting model predictions</i> . <i>Advances in Neural Information Processing Systems</i> . https://proceedings.neurips.cc/paper_files/paper/2017/file/8a20a8621978632d76c43dfd28b67767-Paper.pdf
11	Holzinger, A., Biemann, C., Pattichis, C. S., & Kell, D. B. (2017). <i>What do we need to build explainable AI systems for the medical domain?</i> <i>arXiv preprint arXiv:1712.09923</i> . https://arxiv.org/pdf/1712.09923.pdf
12	Ribeiro, M. T., Singh, S., & Guestrin, C. (2016). “Why should I trust you?”: Explaining the predictions of any classifier. <i>Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining</i> (pp. 1135–1144). https://doi.org/10.1145/2939672.2939778